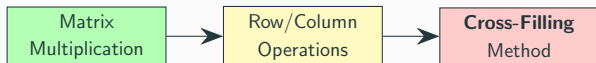


Cross-Filling Method

Finding Hidden Patterns in Recipe Tables

Introduction

Learning Objectives



Key Questions:

- Can we **discover hidden intermediate products** from a direct recipe table?
- What is a **simple pattern** (rank-one table)?
- How to **decompose** complex tables into simple patterns?
- How to convert between **sum** and **product** forms?

Review: Sum-of-Rank-One View

Recall from Lecture 1, matrix multiplication $C = AB$ has three views:

$$C = \begin{pmatrix} | & | & \cdots & | \\ a_1 & a_2 & \cdots & a_n \\ | & | & \cdots & | \end{pmatrix} \begin{pmatrix} - & b_1^T & - \\ - & b_2^T & - \\ \vdots & \vdots & \vdots \\ - & b_n^T & - \end{pmatrix} = \sum_{k=1}^n a_k b_k^T$$

Each $a_k b_k^T$ is called a **rank-one piece** (column $\times$ row).

Today's question: Can we **reverse** this? Given C , find the pieces?

Discovering Simple Patterns (Concretization)

The Motivating Problem

Coffee shop scenario:

We have a **direct** recipe table from raw materials to final products:

			
$C =$		2	2
		1	1
		0	4
		2	1

Question: Are there **hidden intermediate products** (like  ,  , ) that we could use to simplify this?

What is a Simple Pattern?

Look at the first two rows of our table:

		
	2	2
	1	1

Notice: Both products use **the same amount!**

- : 2  + 1 
- : 2  + 1 

Can we extract this **common base** and see what's different?

Setup: The Full Direct Table

		
$C =$ 	2	2
	1	1
	0	4
	2	1

We noticed: first two rows have the same pattern.

Can we **extract** this pattern and see what's left?

Change: Choose a Pivot

Pick a pivot (center of our cross):

Let's pick the 🌿 row and 🍱 column. Entry = 2.

		
	2	2
	1	1
	0	4
	2	1

This pivot will be the **center of our cross**.

Calculation Step 1: Extract the Cross

The **cross** through the pivot includes:

- Entire 🌿 row (row through pivot)
- Entire 🍱 column (column through pivot)

		
	2	2
	1	1
	0	4
	2	1

Cross row: (2, 2) Cross column: (2, 1, 0, 2)

Calculation Step 2: Build the Pattern Table

Idea: Use the cross to fill all entries via "multiplication table":

$$\text{New entry} = \frac{(\text{column value}) \times (\text{row value})}{\text{pivot}}$$

For  and :

$$\text{Entry} = \frac{1 \times 2}{2} = 1$$

For  and :

$$\text{Entry} = \frac{0 \times 2}{2} = 0$$

Result: The First Pattern R_1

		
$R_1 =$		
	2	2
	1	1
	0	0
	2	2

Interpretation: This is the **common base recipe**!

- Base recipe: 2  + 1  + 0  + 2 
-  uses: 1 $\times$ base
-  uses: 1 $\times$ base (+ adjustments)

Insight: What is This Pattern?

The pattern R_1 is a **rank-one matrix**:

$$R_1 = \begin{pmatrix} 2 \\ 1 \\ 0 \\ 2 \end{pmatrix} \begin{pmatrix} 1 & 1 \end{pmatrix}$$

- **Column** (2, 1, 0, 2): the base recipe - **Row** (1, 1): both products use 1× this base

Key: **One pattern, repeated!** Both columns are the same.

Setup: What's Left After R_1 ?

Subtract R_1 from the original table C :

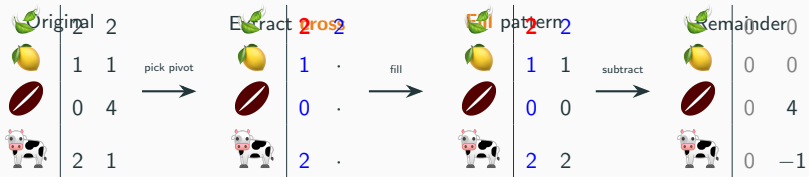
		
$C - R_1 =$ 	$2 - 2$	$2 - 2$
	$1 - 1$	$1 - 1$
	$0 - 0$	$4 - 0$
	$2 - 2$	$1 - 2$

Result: The Remainder

		
$C_2 = C - R_1 =$		
	0	0
	0	0
	0	4
	0	-1

What's left? -  column: all zeros (base recipe exactly matches!) -
 column: adjustments needed (+4  , -1 )

Insight: Why "Cross-Filling"?



The method **fills in** the cross region, then eliminates it!

Change: Find Second Pattern

Look at the remainder:

$$C_2 = \begin{array}{c|cc} & \text{🍱} & \text{🍜} \\ \hline \text{🍫} & 0 & 4 \\ \text{🐮} & 0 & -1 \end{array}$$

This is the **adjustment** for 🍜: $+4$ 🍫, -1 🐮

Pick pivot: 🍫 row, 🍜 column. Entry = 4.

Calculation: Fill Second Cross

Pivot at $(3,2) = 4$. Cross: column $(0, 0, 4, -1)^T$, row $(0, 4)$

$$R_2 = \frac{1}{\textcolor{red}{4}} \begin{pmatrix} 0 \\ 0 \\ 4 \\ -1 \end{pmatrix} \begin{pmatrix} 0 & 4 \end{pmatrix} =$$

		
	0	0
	0	0
	0	4
	0	-1

Interpretation: The **adjustment term** — only affects .

Result: Complete Decomposition

$$C_2 - R_2 =$$

		
	0	0
	0	0

Done! Remainder is zero.

The Full Decomposition

We discovered:

$$C = R_1 + R_2$$

	2	2	=		2	2	+		0	0
	1	1			1	1			0	0
	0	4			0	0			0	4
	2	1			2	2			0	-1

Two **rank-one patterns**: base recipe + adjustment!

Insight: Base + Adjustment Decomposition

What we found:

$$\text{🍲} = 1 \times (\text{base recipe}) + \text{adjustment}$$

Key insight:

- R_1 : Common base shared by both products
- R_2 : Adjustment term (only for 🍲)

- 🍱 = exactly the base recipe

- 🍲 = base + extra 4 🍷 - 1 🐮

This is **expressing complex recipes in terms of simpler patterns!**

Change of Perspective

Change of Perspective

What we've learned so far:

- **What** a simple pattern (rank-one) looks like — one base mix used in different amounts
- **How** to extract patterns using cross-filling — pick pivot, fill cross, subtract
- **Why** this works — discovers hidden intermediate products

From This Point Forward

We **forget about coffee shops** and work with **abstract matrices**:

- **What**: Matrices A , B , C , rank-one pieces R_k
- **How**: Cross-filling algorithm, sum $\leftrightarrow$ product conversion
- **Why**: Compute rank, factorizations, solve equations

This simplifies notation while preserving the intuition.

Rank-One Matrices (Formalization)

Definition: Rank-One Matrix

Definition (Rank-One Matrix)

A matrix R is **rank-one** if it can be written as

$$R = \mathbf{u} \mathbf{v}^T$$

for some nonzero column vector $\mathbf{u}$ and nonzero row vector $\mathbf{v}^T$.

Example:

$$R = \begin{pmatrix} 2 \\ 1 \\ 3 \end{pmatrix} \begin{pmatrix} 1 & 2 \end{pmatrix} = \begin{pmatrix} 2 & 4 \\ 1 & 2 \\ 3 & 6 \end{pmatrix}$$

Key: All columns are multiples of $\mathbf{u}$. All rows are multiples of $\mathbf{v}^T$.

Recognizing Rank-One Structure

Is this matrix rank-one?

$$\begin{pmatrix} 6 & 9 & 3 \\ 4 & 6 & 2 \\ 10 & 15 & 5 \end{pmatrix}$$

Check rows:

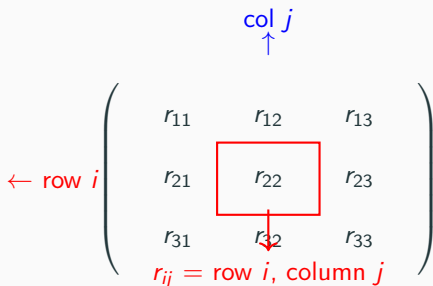
- Row 1: $(6, 9, 3) = 3 \cdot (2, 3, 1)$
- Row 2: $(4, 6, 2) = 2 \cdot (2, 3, 1)$
- Row 3: $(10, 15, 5) = 5 \cdot (2, 3, 1)$

Yes! $R = \begin{pmatrix} 3 \\ 2 \\ 5 \end{pmatrix} \begin{pmatrix} 2 & 3 & 1 \end{pmatrix}$

Cross-Product Property

In a rank-one matrix $R = \mathbf{u}\mathbf{v}^T$, every entry is $r_{ij} = u_i v_j$.

Visual meaning of subscripts:



A 3x3 matrix R is shown with entries r_{11}, r_{12}, r_{13} in the first row, r_{21}, r_{22}, r_{23} in the second row, and r_{31}, r_{32}, r_{33} in the third row. A red arrow points to the second row, labeled "row i ". A blue arrow points to the second column, labeled "col j ". A red rectangle highlights the entry r_{22} . A red arrow points down from r_{22} to the label $r_{ij} = \text{row } i, \text{ column } j$.

$$\left(\begin{array}{ccc} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{array} \right)$$

$r_{ij} = \text{row } i, \text{ column } j$

For any four entries forming a rectangle:

$$r_{ij} \cdot r_{kl} = r_{il} \cdot r_{kj}$$



A partial diagram of a matrix with entries $\cdot$ in the first row. A red rectangle highlights a portion of the matrix.

$$\left(\begin{array}{ccc} \cdot & \cdot & \cdot \end{array} \right)$$

Not Every Matrix is Rank-One

$$C = \begin{pmatrix} 2 & 4 \\ 1 & 2 \\ 3 & 7 \end{pmatrix}$$

Check ratios: $\frac{4}{2} = 2$, $\frac{2}{1} = 2$, $\frac{7}{3} \neq 2$ $\leftarrow$ Different!

This is **not rank-one**. But can we write it as a **sum** of rank-one pieces?

$$C = \underbrace{\begin{pmatrix} ? \\ ? \\ ? \end{pmatrix}}_{R_1} + \underbrace{\begin{pmatrix} ? \\ ? \\ ? \end{pmatrix}}_{R_2} + \cdots$$

Cross-filling answers this!

Cross-Filling Algorithm

The Cross-Filling Algorithm

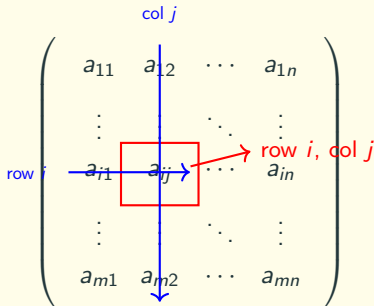
Proposition (Cross-Filling Algorithm)

Input: Matrix A ($m \times n$)

Output: $A = R_1 + R_2 + \cdots + R_r$ (sum of rank-one matrices)

Procedure:

1. Find any nonzero entry $a_{ij} \neq 0$ (the **pivot**)



2. Extract the cross: row i and column j
3. Form rank-one piece:

Pivot Selection Principles

Key Question: Which nonzero entry should we pick as pivot?

Pivot Independence Principle

At each step, choose any nonzero entry $a_{ij} \neq 0$ such that:

- Row i is **not yet zeroed** by previous steps
- Column j is **not yet zeroed** by previous steps

Intuition: Row and column projections must be **independent** (not overlapping with already-eliminated crosses).

Consequence: **Many different pivot sequences are valid!**

- Diagonal pivots $(1, 1) \rightarrow (2, 2) \rightarrow \dots \leftarrow$ one choice
- Off-diagonal pivots $(1, 3) \rightarrow (2, 1) \rightarrow \dots \leftarrow$ also valid!
- Different sequences give different decompositions, but **same rank**

Example: Diagonal Cross-Filling

Decompose using diagonal pivots:

$$A = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 3 & 3 \\ 1 & 3 & 8 \end{pmatrix}$$

Strategy: Pick pivots $(1, 1) \rightarrow (2, 2) \rightarrow (3, 3)$

We will see this produces **LU decomposition** structure.

Iteration 1: Pivot $a_{11} = 1$

Choosing position (1, 1):

col 1

row 1

$$\begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}$$

A red square highlights the element a_{11} . A blue cross is drawn through the first row and first column, with arrows at the ends.

Highlight the cross:

$$A = \begin{pmatrix} \mathbf{1} & 1 & 1 \\ 1 & 3 & 3 \\ 1 & 3 & 8 \end{pmatrix}$$

Fill using the cross:

$$1 \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \begin{pmatrix} 1 & 1 & 1 \end{pmatrix}$$

Iteration 1: Remainder A_1

$$A_1 = A - R_1 = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 3 & 3 \\ 1 & 3 & 8 \end{pmatrix} - \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 2 & 2 \\ 0 & 2 & 7 \end{pmatrix}$$

Row 1 and column 1 are now zero. ✓

Iteration 2: Pivot $(A_1)_{22} = 2$

Choosing position (2, 2) in the remainder:

col 2

row 2

$$\begin{pmatrix} 0 & 0 & 0 \\ 0 & a_{22} & a_{23} \\ 0 & a_{32} & a_{33} \end{pmatrix}$$

^

$$A_1 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 2 & 2 \\ 0 & 2 & 7 \end{pmatrix}$$

Fill:

$$P = \frac{1}{2} \begin{pmatrix} 0 \\ 2 \\ 2 \end{pmatrix} \begin{pmatrix} 0 & 2 & 2 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 2 & 2 \\ 0 & 2 & 2 \end{pmatrix}$$

Iteration 2: Remainder A_2

$$A_2 = A_1 - R_2 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 2 & 2 \\ 0 & 2 & 7 \end{pmatrix} - \begin{pmatrix} 0 & 0 & 0 \\ 0 & 2 & 2 \\ 0 & 2 & 2 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 5 \end{pmatrix}$$

Rows 1-2 and columns 1-2 are now zero. ✓

Iteration 3: Final Piece

$$A_2 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & \mathbf{5} \end{pmatrix}$$

This is already rank-one:

$$R_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \begin{pmatrix} 0 & 0 & 5 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & \mathbf{5} \end{pmatrix}$$

Remainder: $A_3 = A_2 - R_3 = O$. **Done!**

Full Decomposition (Diagonal Pivots)

$$A = \underbrace{\begin{pmatrix} \color{red}{1} & \color{blue}{1} & \color{blue}{1} \\ \color{blue}{1} & 1 & 1 \\ \color{blue}{1} & 1 & 1 \end{pmatrix}}_{R_1} + \underbrace{\begin{pmatrix} 0 & 0 & 0 \\ 0 & \color{red}{2} & \color{blue}{2} \\ 0 & \color{blue}{2} & 2 \end{pmatrix}}_{R_2} + \underbrace{\begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & \color{red}{5} \end{pmatrix}}_{R_3}$$

Three rank-one pieces:

- R_1 : pivot **1** at position (1, 1)
- R_2 : pivot **2** at position (2, 2)
- R_3 : pivot **5** at position (3, 3)

Key: **Diagonal** pivot strategy $\rightarrow$ special structure (LU decomposition)

Example: Non-Diagonal Cross-Filling (Setup)

Now let's decompose the **same matrix** using **off-diagonal pivots**:

$$A = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 3 & 3 \\ 1 & 3 & 8 \end{pmatrix}$$

New strategy: Pick pivots $(1, 3) \rightarrow (2, 1) \rightarrow (3, 2)$ (all off-diagonal!)

Question: Will we get a different decomposition?

Answer: Yes! Different R_k pieces, but **same rank** (still 3 pieces).

Non-Diagonal Iteration 1: Pivot (1, 3)

Choose position (1, 3): entry = 1

$$A = \begin{pmatrix} 1 & 1 & \mathbf{1} \\ 1 & 3 & \mathbf{3} \\ 1 & 3 & \mathbf{8} \end{pmatrix} \quad \text{Cross: row 1} = (\mathbf{1}, \mathbf{1}, \mathbf{1}), \text{ col 3} = (\mathbf{1}, \mathbf{3}, \mathbf{8})^T$$

Fill:

$$R'_1 = \frac{1}{\mathbf{1}} \begin{pmatrix} \mathbf{1} \\ \mathbf{3} \\ \mathbf{8} \end{pmatrix} (\mathbf{1} \quad \mathbf{1} \quad \mathbf{1}) = \begin{pmatrix} \mathbf{1} & \mathbf{1} & \mathbf{1} \\ \mathbf{3} & \mathbf{3} & \mathbf{3} \\ \mathbf{8} & \mathbf{8} & \mathbf{8} \end{pmatrix}$$

Remainder:

$$A'_1 = A - R'_1 = \begin{pmatrix} 0 & 0 & 0 \\ -2 & 0 & 0 \\ -7 & -5 & 0 \end{pmatrix}$$

Row 1 and column 3 are zero. ✓

Non-Diagonal Iteration 2: Pivot (2, 1)

Choose position (2, 1) in A'_1 : entry = -2

$$A'_1 = \begin{pmatrix} 0 & 0 & 0 \\ -2 & 0 & 0 \\ -7 & -5 & 0 \end{pmatrix}$$

Fill:

$$R'_2 = \frac{1}{-2} \begin{pmatrix} 0 \\ -2 \\ -7 \end{pmatrix} \begin{pmatrix} -2 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 \\ -2 & 0 & 0 \\ -7 & 0 & 0 \end{pmatrix}$$

Remainder:

$$A'_2 = A'_1 - R'_2 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & -5 & 0 \end{pmatrix}$$

Non-Diagonal Iteration 3: Final Piece

$$A'_2 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & -5 & 0 \end{pmatrix}$$

$$R'_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \begin{pmatrix} 0 & -5 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & -5 & 0 \end{pmatrix}$$

Done! $A = R'_1 + R'_2 + R'_3$

Comparison: Diagonal vs Non-Diagonal

Same matrix A , different pivot strategies:

	Diagonal $(1, 1) \rightarrow (2, 2) \rightarrow (3, 3)$	Off-diagonal $(1, 3) \rightarrow (2, 1) \rightarrow (3, 2)$
R_1	$\begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}$	$\begin{pmatrix} 1 & 1 & 1 \\ 3 & 3 & 3 \\ 8 & 8 & 8 \end{pmatrix}$
R_2	$\begin{pmatrix} 0 & 0 & 0 \\ 0 & 2 & 2 \\ 0 & 2 & 2 \end{pmatrix}$	$\begin{pmatrix} 0 & 0 & 0 \\ -2 & 0 & 0 \\ -7 & 0 & 0 \end{pmatrix}$
R_3	$\begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 5 \end{pmatrix}$	$\begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & -5 & 0 \end{pmatrix}$
Rank	3	3

Key: Different decompositions, same rank!

Another Example: 3×4 Matrix

Decompose with non-diagonal pivots:

$$B = \begin{pmatrix} 2 & 0 & 4 & 2 \\ 1 & 3 & 5 & 4 \\ 0 & 6 & 2 & 5 \end{pmatrix}$$

Strategy: $(2, 3) \rightarrow (1, 1) \rightarrow (3, 2)$

Step 1: Pivot at $(2, 3)$, entry = 5

Cross: row 2 = $(1, 3, 5, 4)$, col 3 = $(4, 5, 2)^T$

$$R_1 = \frac{1}{5} \begin{pmatrix} 4 \\ 5 \\ 2 \end{pmatrix} \begin{pmatrix} 1 & 3 & 5 & 4 \end{pmatrix} = \begin{pmatrix} 4/5 & 12/5 & 4 & 16/5 \\ 1 & 3 & 5 & 4 \\ 2/5 & 6/5 & 2 & 8/5 \end{pmatrix}$$

3×4 Remainder After Step 1

$$B_1 = B - R_1 = \begin{pmatrix} 6/5 & -12/5 & 0 & -6/5 \\ 0 & 0 & 0 & 0 \\ -2/5 & 24/5 & 0 & 17/5 \end{pmatrix}$$

Row 2 and column 3 are zero.

Step 2: Pivot at $(1, 1)$, entry = $6/5$

$$R_2 = \frac{5}{6} \begin{pmatrix} 6/5 \\ 0 \\ -2/5 \end{pmatrix} \begin{pmatrix} 6/5 & -12/5 & 0 & -6/5 \end{pmatrix} = \begin{pmatrix} 6/5 & -12/5 & 0 & -6/5 \\ 0 & 0 & 0 & 0 \\ -2/5 & 4/5 & 0 & 2/5 \end{pmatrix}$$

$$B_2 = B_1 - R_2 = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 4 & 0 & 3 \end{pmatrix}$$

3 × 4 Final Step

$$B_2 = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 4 & 0 & 3 \end{pmatrix}$$

Step 3: Pivot at $(3, 2)$, entry = 4

$$R_3 = \frac{1}{4} \begin{pmatrix} 0 \\ 0 \\ 4 \end{pmatrix} \begin{pmatrix} 0 & 4 & 0 & 3 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 4 & 0 & 3 \end{pmatrix}$$

$$B_3 = B_2 - R_3 = O. \quad \text{Done!}$$

Result: $B = R_1 + R_2 + R_3$, $\text{rank}(B) = 3$

Key Insight: Pivot Flexibility

Main Takeaway

Pivot selection is flexible!

- At each step, choose **any nonzero entry** in a non-zeroed row and column
- Different pivot choices $\rightarrow$ different rank-one pieces R_k
- But the **total number of pieces** (= rank) is always the same

Why does this matter?

- **Computational:** Choose pivots to avoid small numbers (numerical stability)
- **Structural:** Diagonal pivots $\rightarrow$ LU decomposition
- **Theoretical:** Rank is well-defined (independent of decomposition method)

Next lecture: Prove that rank is well-defined via linear independence.

Definition: Rank of a Matrix

Definition (Rank)

The **rank** of a matrix A is the number of rank-one pieces in its cross-filling decomposition:

$$A = R_1 + R_2 + \cdots + R_r \implies \text{rank}(A) = r$$

Examples:

- Zero matrix O : rank 0
- Rank-one matrix $\mathbf{uv}^T$: rank 1
- Identity I_n : rank n
- Our A above: rank 3 (both decompositions gave 3 pieces)
- The 3×4 matrix B : rank 3

Important: Different pivot choices give different decompositions, but always the same count r . (Proved in Lecture 4.)

Sum $\Leftrightarrow$ Product Equivalence

From Sum to Product Form

Each rank-one piece is an **outer product**:

$$R_k = \mathbf{u}_k \mathbf{v}_k^T$$

From our diagonal example:

$$R_1 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \begin{pmatrix} 1 & 1 & 1 \end{pmatrix}$$

$$R_2 = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} \begin{pmatrix} 0 & 2 & 2 \end{pmatrix}$$

$$R_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \begin{pmatrix} 0 & 0 & 5 \end{pmatrix}$$

Collecting the Pieces

Collect columns $\mathbf{u}_k$ into U , rows $\mathbf{v}_k^T$ into V :

$$U = \begin{pmatrix} | & | & | \\ \mathbf{u}_1 & \mathbf{u}_2 & \mathbf{u}_3 \\ | & | & | \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{pmatrix}$$

$$V = \begin{pmatrix} - & \mathbf{v}_1^T & - \\ - & \mathbf{v}_2^T & - \\ - & \mathbf{v}_3^T & - \end{pmatrix} = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 2 & 2 \\ 0 & 0 & 5 \end{pmatrix}$$

By sum-of-rank-one: $UV = \sum_{k=1}^3 \mathbf{u}_k \mathbf{v}_k^T = R_1 + R_2 + R_3 = A$

Verification

$$UV = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 & 1 \\ 0 & 2 & 2 \\ 0 & 0 & 5 \end{pmatrix} = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 3 & 3 \\ 1 & 3 & 8 \end{pmatrix} = A \quad \checkmark$$

Notice: U is **lower triangular**, V is **upper triangular**!

This happened because we chose **diagonal pivots**
 $(1, 1) \rightarrow (2, 2) \rightarrow (3, 3)$.

Key: Diagonal pivots $\implies$ **LU decomposition** (preview of next section).

Sum $\leftrightarrow$ Product Equivalence

Proposition (Sum $\leftrightarrow$ Product)

For any matrix A :

$$A = \sum_{k=1}^r \mathbf{u}_k \mathbf{v}_k^T \iff A = U \cdot V$$

where U has columns $\mathbf{u}_k$ and V has rows $\mathbf{v}_k^T$.

Two directions:

- **Lecture 1:** Given product $A = UV$, read off sum $A = \sum \mathbf{u}_k \mathbf{v}_k^T$
- **Today:** Given sum from cross-filling, assemble product $A = UV$

The Inner Dimension

When $A = UV$ with A being $m \times n$:

$$A_{m \times n} = U_{m \times \boxed{r}} \cdot V_{\boxed{r} \times n}$$

The shared dimension r is the **inner dimension**.

- Outer dimensions m, n : visible (size of A)
- Inner dimension r : **hidden** = number of independent patterns

Key: $\text{rank}(A) = r = \text{inner dimension of the factorization}$.

LU Decomposition via Diagonal Cross-Filling

What is LU Decomposition?

Goal: Write $A = LU$ where:

- L is **lower triangular** with **diagonal = 1**
- U is **upper triangular**

Example:

$$\begin{pmatrix} 2 & 4 & 6 \\ 1 & 5 & 7 \\ 3 & 9 & 10 \end{pmatrix} = \underbrace{\begin{pmatrix} 1 & 0 & 0 \\ \ell_{21} & 1 & 0 \\ \ell_{31} & \ell_{32} & 1 \end{pmatrix}}_L \underbrace{\begin{pmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{pmatrix}}_U$$

Key constraint: L must have 1's on the diagonal!

Method: Use **diagonal cross-filling** with a special technique.

The Key Technique: Normalizing L's Diagonal

Recall the cross-filling formula at pivot (k, k) :

$$R_k = \frac{1}{a_{kk}} \cdot (\text{column } k) \cdot (\text{row } k)$$

Rewrite to separate the normalization:

$$R_k = \underbrace{\left(\frac{\text{column } k}{a_{kk}} \right)}_{\text{normalized column}} \cdot (\text{row } k)$$

Key observation:

- The k -th entry of the normalized column $= \frac{a_{kk}}{a_{kk}} = 1$
- Collect these normalized columns $\rightarrow L$ has diagonal $= 1$!
- Collect the original rows $\rightarrow U$ (upper triangular)

This is the **trick** that ensures L has diagonal $= 1$.

Example: 4×4 Matrix LU Decomposition

Decompose via diagonal cross-filling:

$$A = \begin{pmatrix} 2 & 4 & -2 & 2 \\ 4 & 9 & -3 & 5 \\ -2 & -3 & 7 & -1 \\ 2 & 5 & -1 & 6 \end{pmatrix}$$

Strategy: Diagonal pivots $(1, 1) \rightarrow (2, 2) \rightarrow (3, 3) \rightarrow (4, 4)$

We will extract R_1, R_2, R_3, R_4 and assemble L and U .

Step 1: Pivot at (1, 1), entry = 2

$$A = \begin{pmatrix} \color{red}{2} & \color{blue}{4} & \color{blue}{-2} & \color{blue}{2} \\ \color{blue}{4} & 9 & -3 & 5 \\ \color{blue}{-2} & -3 & 7 & -1 \\ \color{blue}{2} & 5 & -1 & 6 \end{pmatrix}$$

Cross: column = $(2, 4, -2, 2)^T$, row = $(2, 4, -2, 2)$

Apply the technique:

$$R_1 = \underbrace{\frac{1}{2} \begin{pmatrix} 2 \\ 4 \\ -2 \\ 2 \end{pmatrix}}_{= \text{column 1 of } L} \cdot \underbrace{\begin{pmatrix} 2 & 4 & -2 & 2 \end{pmatrix}}_{= \text{row 1 of } U} = \underbrace{\begin{pmatrix} 1 \\ 2 \\ -1 \\ 1 \end{pmatrix}}_{\ell_1} \begin{pmatrix} 2 & 4 & -2 & 2 \end{pmatrix}$$

Note: ℓ_1 has first entry = 1

Remainder After Step 1

$$A_1 = A - R_1 = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 \\ 0 & 1 & 5 & 1 \\ 0 & 1 & 1 & 4 \end{pmatrix}$$

Row 1 and column 1 are zero.

Next: Pivot at (2,2) in the non-zero block.

Step 2: Pivot at $(2, 2)$, entry = 1

$$A_1 = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & \mathbf{1} & \mathbf{1} & \mathbf{1} \\ 0 & \mathbf{1} & 5 & 1 \\ 0 & \mathbf{1} & 1 & 4 \end{pmatrix}$$

Cross: column = $(0, 1, 1, 1)^T$, row = $(0, 1, 1, 1)$

Apply the technique:

$$R_2 = \underbrace{\frac{1}{1} \begin{pmatrix} 0 \\ 1 \\ 1 \\ 1 \end{pmatrix}}_{= \text{column 2 of } L} \cdot \underbrace{\begin{pmatrix} 0 & 1 & 1 & 1 \end{pmatrix}}_{= \text{row 2 of } U} = \underbrace{\begin{pmatrix} 0 \\ 1 \\ 1 \\ 1 \end{pmatrix}}_{\ell_2} \begin{pmatrix} 0 & 1 & 1 & 1 \end{pmatrix}$$

Note: ℓ_2 has second entry = 1

Remainder After Step 2

$$A_2 = A_1 - R_2 = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 4 & 0 \\ 0 & 0 & 0 & 3 \end{pmatrix}$$

Rows 1–2 and columns 1–2 are zero.

Next: Pivot at (3,3).

Step 3: Pivot at (3,3), entry = 4

$$A_2 = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & \mathbf{4} & \mathbf{0} \\ 0 & 0 & \mathbf{0} & \mathbf{3} \end{pmatrix}$$

Cross: column = $(0, 0, 4, 0)^T$, row = $(0, 0, 4, 0)$

Apply the technique:

$$R_3 = \underbrace{\frac{1}{4} \begin{pmatrix} 0 \\ 0 \\ 4 \\ 0 \end{pmatrix}}_{= \text{column 3 of } L} \cdot \underbrace{\begin{pmatrix} 0 & 0 & 4 & 0 \end{pmatrix}}_{= \text{row 3 of } U} = \underbrace{\begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}}_{\ell_3} \begin{pmatrix} 0 & 0 & 4 & 0 \end{pmatrix}$$

Note: ℓ_3 has third entry = 1

Step 4: Final Pivot at (4,4), entry = 3

$$A_3 = A_2 - R_3 = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \mathbf{3} \end{pmatrix}$$

$$R_4 = \underbrace{\frac{1}{3} \begin{pmatrix} 0 \\ 0 \\ 0 \\ 3 \end{pmatrix}}_{= \text{column 4 of } L} \cdot \underbrace{\begin{pmatrix} 0 & 0 & 0 & 3 \end{pmatrix}}_{= \text{row 4 of } U} = \underbrace{\begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}}_{\ell_4} \begin{pmatrix} 0 & 0 & 0 & 3 \end{pmatrix}$$

Note: ℓ_4 has fourth entry = 1

$$A_4 = A_3 - R_4 = O. \quad \text{Done!}$$

Assembling L and U

Collect normalized columns into L :

$$L = \begin{pmatrix} | & | & | & | \\ \ell_1 & \ell_2 & \ell_3 & \ell_4 \\ | & | & | & | \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 2 & 1 & 0 & 0 \\ -1 & 1 & 1 & 0 \\ 1 & 1 & 0 & 1 \end{pmatrix}$$

Key: Diagonal = $(1, 1, 1, 1)$

Collect rows into U :

$$U = \begin{pmatrix} - & \text{row 1} & - \\ - & \text{row 2} & - \\ - & \text{row 3} & - \\ - & \text{row 4} & - \end{pmatrix} = \begin{pmatrix} 2 & 4 & -2 & 2 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 4 & 0 \\ 0 & 0 & 0 & 3 \end{pmatrix}$$

Key: Upper triangular, pivots on diagonal

Verification: $LU = A$

$$LU = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 2 & 1 & 0 & 0 \\ -1 & 1 & 1 & 0 \\ 1 & 1 & 0 & 1 \end{pmatrix} \begin{pmatrix} 2 & 4 & -2 & 2 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 4 & 0 \\ 0 & 0 & 0 & 3 \end{pmatrix}$$

First row: $(1)(2, 4, -2, 2) = (2, 4, -2, 2)$

Second row:

$$(2)(2, 4, -2, 2) + (1)(0, 1, 1, 1) = (4, 8, -4, 4) + (0, 1, 1, 1) = (4, 9, -3, 5)$$

$$\begin{aligned} \text{Third row: } & (-1)(2, 4, -2, 2) + (1)(0, 1, 1, 1) + (1)(0, 0, 4, 0) = \\ & (-2, -4, 2, -2) + (0, 1, 1, 1) + (0, 0, 4, 0) = (-2, -3, 7, -1) \end{aligned}$$

Fourth row:

$$\begin{aligned} & (1)(2, 4, -2, 2) + (1)(0, 1, 1, 1) + (0)(0, 0, 4, 0) + (1)(0, 0, 0, 3) = \\ & (2, 5, -1, 6) \end{aligned}$$

$$LU = A \quad \begin{pmatrix} 2 & 4 & -2 & 2 \\ 4 & 9 & -3 & 5 \\ -2 & -3 & 7 & -1 \\ 2 & 5 & -1 & 6 \end{pmatrix}$$

Why Does This Work?

The magic: By normalizing each column by its pivot, we ensure:

$$\ell_k = \frac{\text{column } k}{a_{kk}} \implies (\ell_k)_k = \frac{a_{kk}}{a_{kk}} = 1$$

In general:

- Cross-filling with diagonal pivots: $A = R_1 + R_2 + \cdots + R_n$
- Each $R_k = \ell_k \cdot (\text{row } k)$ where $\ell_k = (\text{col } k)/a_{kk}$
- Collect: $L = [\ell_1 \mid \ell_2 \mid \cdots \mid \ell_n]$, rows into U
- Result: $A = LU$ with L lower triangular (diagonal = 1), U upper triangular

Summary:

Diagonal cross-filling + normalization trick = LU decomposition

When Diagonal LU Fails

Diagonal cross-filling requires **nonzero diagonal pivots**. But this can fail:

$$A = \begin{pmatrix} 0 & 3 \\ 1 & 2 \end{pmatrix}$$

Problem: position (1,1)

$$\begin{pmatrix} \boxed{a_{11}} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}$$

Pivot $a_{11} = 0$ — cannot start!

Even later pivots can fail:

$$\begin{pmatrix} 1 & 2 & 3 \\ 1 & 2 & 4 \\ 1 & 3 & 5 \end{pmatrix} \xrightarrow{\text{after } R_1} \begin{pmatrix} 0 & 0 & 0 \\ 0 & \mathbf{0} & 1 \\ 0 & 1 & 2 \end{pmatrix}$$

Summary

Summary

Today we learned:

1. **Rank-one matrix:** $R = \mathbf{u} \mathbf{v}^T$ (one pattern, repeated)
2. **Cross-filling algorithm:** Decompose any matrix into rank-one pieces

$$A = R_1 + R_2 + \cdots + R_r$$

Pick pivot $\rightarrow$ Extract cross $\rightarrow$ Fill $\rightarrow$ Subtract $\rightarrow$ Repeat

3. **Pivot selection principle:** Choose any nonzero entry in independent row/column
4. **Pivot flexibility:** Different pivot sequences $\rightarrow$ different decompositions, same rank
5. **Sum $\leftrightarrow$ Product equivalence:**

$$\sum_{k=1}^r \mathbf{u}_k \mathbf{v}_k^T = A = UV$$

6. **Rank** = number of pieces = inner dimension of UV
7. **Special case:** Diagonal pivots $\rightarrow$ LU decomposition

Lecture 4 (next week):

- **Well-definedness of rank** — Why does rank not depend on pivot choices?
- **Linear independence** and **basis** — Two languages for subspaces
- **Dimension** — Connection to rank via trace
- Solving $Ax = b$ using cross-filling

The big picture:

Cross-filling $\rightarrow$ Rank $\rightarrow$ Subspaces $\rightarrow$ Projections $\rightarrow$ Spectral
Decomposition

Every major topic connects back to rank-one decomposition!